Lesson Planning

Complete Lesson Plan Book

Class: Class 7

Subject: Science

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: The Cell: Building Block of Life

Learning Objectives

• Define a cell and state the cell theory.

• Label the parts of plant and animal cells.

• Compare plant and animal cells.

• Explain the functions of the main organelles.

Key Concepts

• All living things are made of cells; the cell is the basic unit of life.

• Cells were first observed by Robert Hooke in 1665.

• Plant cells have a cell wall and chloroplasts; animal cells do not.

• The nucleus controls the cell; mitochondria release energy.

• The cell membrane controls what enters and leaves the cell.

Teaching Methodology (Step by Step)

1. Show the class a brick wall and compare it to an organism made of cells.
2. Draw and label a plant cell and an animal cell on the board.
3. Explain each organelle with a simple function analogy (nucleus = control room).
4. Students complete a Venn diagram comparing the two cell types.
5. Quick oral quiz: point to a part on the chart and ask its name and function.

Classroom Activities

• Build a 3D cell model using clay, beads and a plastic container.

• Observe onion peel cells under the microscope in the laboratory.

Assessment Questions

1. Name three differences between plant and animal cells.
2. What is the function of the cell wall and the chloroplast?
3. Draw and label an animal cell.

Homework

• Make a labelled chart of a plant cell and write one function for each part.

Time Allocation

Introduction 5 min, teaching 20 min, diagram practice 10 min, quiz 5 min.

Unit 2: Human Body Systems

Learning Objectives

• Name the major systems of the human body.

• Describe the path of food through the digestive system.

• Explain how the heart pumps blood around the body.

• State the role of the skeleton and muscles.

Key Concepts

• Organs working together form a system.

• Digestion breaks food into tiny particles the body can absorb.

• The heart is a muscular pump with four chambers.

• Blood carries oxygen, food and waste around the body.

• The skeleton gives shape and protection; muscles move the bones.

Teaching Methodology (Step by Step)

1. Ask students to list what their body does in one minute (breathe, heartbeat).
2. Present a body chart and name the digestive organs in order.
3. Trace the journey of a sandwich from mouth to large intestine.
4. Explain the double circulation path with a red/blue diagram.
5. Students label a blank body-systems worksheet in pairs.

Classroom Activities

• Measure pulse rate before and after exercise and record the difference.

• Model the digestive tract using a long tube and sponges.

Assessment Questions

1. List the organs of the digestive system in order.
2. Why does the pulse rate increase after running?
3. Name two functions of the human skeleton.

Homework

• Draw the digestive system and label its parts with one-line functions.

Time Allocation

Warm-up 5 min, teaching 20 min, activity 10 min, worksheet 5 min.

Unit 3: Force and Motion

Learning Objectives

• Define force and list its effects on objects.

• Distinguish between contact and non-contact forces.

• Define speed and calculate it from distance and time.

• Explain friction and its uses in daily life.

Key Concepts

• A force is a push or a pull; it is measured in units of force.

• Forces can change the shape, speed or direction of an object.

• Gravity and magnetism act without touching (non-contact forces).

• Speed = distance ÷ time.

• Friction opposes motion; it helps us walk but wears out machine parts.

Teaching Methodology (Step by Step)

1. Push a table and pull a door to show pushes and pulls.
2. List effects of force: stop, start, change direction, change shape.
3. Classify classroom examples into contact and non-contact forces.
4. Solve speed problems on the board: a car travels 120 km in 2 hours.
5. Discuss friction: rub hands together and list where friction helps or harms.

Classroom Activities

• Roll toy cars on smooth and rough surfaces and compare distances.

• Time students walking 20 metres and calculate each one's speed.

Assessment Questions

1. Give two examples each of contact and non-contact forces.
2. A cyclist covers 45 km in 3 hours. Find her speed.
3. Why are shoe soles made rough?

Homework

• List ten forces you observe at home and classify them.

Time Allocation

Demo 5 min, teaching 20 min, numerical practice 10 min, summary 5 min.

Unit 4: Energy and Its Forms

Learning Objectives

• Define energy and name its main forms.

• Give examples of energy transformation.

• Distinguish between kinetic and potential energy.

• State the law of conservation of energy in simple words.

Key Concepts

• Energy is the ability to do work.

• Forms include kinetic, potential, heat, light, sound and electrical energy.

• Energy changes from one form to another, e.g. a battery lighting a bulb.

• A stretched bow has potential energy; a moving car has kinetic energy.

• Energy is never created or destroyed, only changed in form.

Teaching Methodology (Step by Step)

1. Drop a ball and ask where its energy goes at each stage.
2. List energy forms with one daily-life example for each.
3. Explain kinetic vs potential energy with a swinging pendulum.
4. Trace energy changes in a torch and in a moving bicycle.
5. Students complete an energy-transformation matching worksheet.

Classroom Activities

• Build a simple rubber-band catapult and observe energy change.

• Energy hunt: find and list five energy transformations in the classroom.

Assessment Questions

1. Name the energy changes when a candle burns.
2. Distinguish between kinetic and potential energy with one example each.
3. What happens to the total energy when a ball falls to the ground?

Homework

• Write ten examples of energy transformation seen at home.

Time Allocation

Demo 5 min, teaching 20 min, worksheet 10 min, review 5 min.

Unit 5: Matter and Chemical Symbols

Learning Objectives

• Define element, compound and mixture.

• Write the symbols of the first twenty elements.

• Distinguish between physical and chemical changes.

• State the particle model of matter.

Key Concepts

• Matter is anything that has mass and occupies space.

• An element is made of only one kind of atom.

• Compounds form when elements join chemically in fixed ratios.

• Chemical symbols use one or two letters, e.g. H, O, Na, Ca.

• In a chemical change a new substance forms; in a physical change it does not.

Teaching Methodology (Step by Step)

1. Show samples: iron nail (element), water (compound), air (mixture).
2. Introduce the symbol chart and practise writing symbols.
3. Demonstrate a physical change (melting ice) and a chemical change (burning paper).
4. Explain the particle model: particles of solids, liquids and gases.
5. Rapid-fire symbol quiz using flash cards.

Classroom Activities

• Sort picture cards into elements, compounds and mixtures.

• Flash-card race to name element symbols.

Assessment Questions

1. Write the symbols of sodium, calcium, carbon and iron.
2. Give two differences between a compound and a mixture.
3. Is rusting of iron a physical or chemical change? Why?

Homework

• Learn the symbols of the first 20 elements and write each three times.

Time Allocation

Demo 5 min, teaching 20 min, quiz 10 min, summary 5 min.

Unit 6: Acids, Bases and Salts

Learning Objectives

• Define acids and bases and list their properties.

• Identify common acids and bases in daily life.

• Explain the use of indicators like litmus.

• Describe neutralisation with an example.

Key Concepts

• Acids taste sour; bases taste bitter and feel soapy.

• Litmus turns red in acid and blue in base.

• Vinegar, lemon juice and citrus fruits contain acids.

• Soap, baking soda and toothpaste contain bases.

• Acid + base → salt + water (neutralisation).

Teaching Methodology (Step by Step)

1. Let students safely smell vinegar and taste lemon to introduce acids.
2. List properties of acids and bases in a comparison table.
3. Demonstrate litmus paper changing colour in lemon juice and soap solution.
4. Explain neutralisation using the example of antacid tablets.
5. Students test three household solutions with litmus strips.

Classroom Activities

• Make a natural indicator from red cabbage and test household liquids.

• Classify ten daily-life substances as acid, base or neutral.

Assessment Questions

1. What colour does blue litmus turn in lemon juice?
2. Name two acids and two bases found at home.
3. What are the products of neutralisation?

Homework

• Find five acids and five bases in your kitchen and list them.

Time Allocation

Demo 5 min, teaching 20 min, testing activity 10 min, recap 5 min.

Unit 7: Simple Machines

Learning Objectives

• Define a machine and list the six simple machines.

• Explain how levers, pulleys and inclined planes help us.

• Identify simple machines in everyday tools.

• State that machines reduce effort but not work.

Key Concepts

• A machine is a device that makes work easier.

• The six simple machines are lever, pulley, wheel and axle, inclined plane, wedge and screw.

• A lever has a fulcrum, load and effort.

• Pulleys change the direction of force.

• Machines multiply force or distance but never create energy.

Teaching Methodology (Step by Step)

1. Lift a heavy book directly, then with a ruler as a lever — compare.
2. Introduce each simple machine with a classroom example.
3. Draw and label a first-class lever with fulcrum, load and effort.
4. Show a flag pulley and an inclined-plane (ramp) demo.
5. Students hunt the classroom for examples of each simple machine.

Classroom Activities

• Build a working pulley model with string and a spool.

• Scavenger hunt: photograph or sketch simple machines at home.

Assessment Questions

1. Name the six simple machines.
2. Label the fulcrum, load and effort in a seesaw diagram.
3. Give two examples of inclined planes in daily life.

Homework

• Find one example of each simple machine at home and draw them.

Time Allocation

Demo 5 min, teaching 20 min, model building 10 min, quiz 5 min.

Unit 8: Environment and Ecosystems

Learning Objectives

• Define ecosystem and give examples.

• Explain the roles of producers, consumers and decomposers.

• Draw a simple food chain.

• Suggest ways to protect the environment.

Key Concepts

• An ecosystem is a community of living things plus their surroundings.

• Producers (green plants) make food; consumers eat it.

• Decomposers like bacteria return nutrients to the soil.

• A food chain shows who eats whom.

• Pollution, deforestation and waste harm ecosystems.

Teaching Methodology (Step by Step)

1. Show a picture of a pond and ask what lives there and what they need.
2. Introduce producers, consumers and decomposers with examples.
3. Build a grass → goat → wolf food chain on the board.
4. Discuss local environmental problems: smoke, plastic waste.
5. Students design a poster: 'Save Our Environment'.

Classroom Activities

• Observe the school garden and list one food chain found there.

• Poster competition on keeping the neighbourhood clean.

Assessment Questions

1. Draw a food chain with four organisms.
2. What is the role of decomposers?
3. Suggest three ways students can reduce plastic waste.

Homework

• Write a paragraph on how to keep your street clean and green.

Time Allocation

Discussion 5 min, teaching 20 min, poster work 10 min, sharing 5 min.